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Comprehensive Question Bank: Chapter 6 — Data and Analysis

Grade 12 Computer Science | Complete Assessment Bank (MCQs · SQs · LQs)

Assessment Guide for Teachers

- **Section A** — Multiple-Choice Questions (MCQs): Correct answer is **bolded** within the option list. No separate answer key needed.
 - **Section B** — Short Answer Questions (SQs): No answers provided. Designed for 2–4 sentence responses.
 - **Section C** — Long / Extensive Questions (LQs): No answers provided. Structured with sub-parts for guided essay responses.
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Section A: Multiple-Choice Questions (MCQs)

Topic 6.1: Introduction to Data Science

1. What is the primary goal of data science?
 - A) To design computer hardware components
 - B) To convert raw data into meaningful knowledge and support decision-making**
 - C) To write operating system software
 - D) To manage computer networks
2. Which of the following best describes data science?
 - A) A field focused exclusively on programming
 - B) An interdisciplinary field combining mathematics, statistics, computer science, and domain knowledge**
 - C) A branch of hardware engineering
 - D) A field that studies network security
3. John Snow's cholera map (1854) is historically significant in data science because it demonstrated:
 - A) The invention of the modern computer
 - B) That collecting and mapping data can reveal patterns that solve real-world problems**
 - C) The first use of machine learning algorithms
 - D) The discovery of the internet
4. Which of the following is NOT one of the disciplines combined in data science?
 - A) Mathematics
 - B) Statistics
 - C) Mechanical engineering**
 - D) Computer science
5. The term "interdisciplinary" in data science means:
 - A) The field only uses one type of tool
 - B) The field draws knowledge from multiple different subjects working together**
 - C) The field is only for university students
 - D) The field only applies to the internet

6. What is the FIRST stage in the data science workflow pipeline?
- A) Data collection**
 - B) Modelling and training
 - C) Decision making
 - D) Data analysis
7. Which stage of the data science pipeline involves cleaning and processing raw data?
- A) Data collection
 - B) Data analysis (exploring and cleaning)**
 - C) Modelling
 - D) Feedback loop
8. What does the feedback loop stage of the data science workflow do?
- A) It collects new raw data from scratch
 - B) It uses results from decisions to improve the model over time**
 - C) It trains the first version of the model
 - D) It stores data in the cloud
9. "Data science turns data into knowledge." This statement refers to:
- A) Building websites using data
 - B) Analysing raw data to extract useful patterns and insights**
 - C) Storing data securely on servers
 - D) Transmitting data across networks
10. Which skill is described as essential for explaining data science results clearly to others?
- A) Advanced calculus
 - B) Communication skills**
 - C) Network administration
 - D) Hardware design
11. Data science supports decision-making by:
- A) Replacing all human workers with machines
 - B) Reducing guesswork by basing decisions on facts and patterns found in data**
 - C) Eliminating the need for management in organisations
 - D) Only analysing financial data
12. Which of the following is a real-world area where data science is widely used?
- A) Furniture manufacturing only
 - B) Healthcare, education, and business**
 - C) Architecture and construction only
 - D) Transportation engineering only

13. In the data science workflow, what happens AFTER modelling and training?
- A) Data collection
 - B) Data cleaning
 - C) Evaluation of model performance**
 - D) Feedback loop begins immediately
14. Which role does "domain knowledge" play in data science?
- A) It replaces the need for programming
 - B) It helps understand real-world problems so that analysis is meaningful and relevant**
 - C) It provides the computing hardware
 - D) It writes the machine learning algorithms automatically
15. Arthur Samuel coined the term "Machine Learning" in:
- A) 1839
 - B) 1959**
 - C) 1989
 - D) 2009
-

Topic 6.2: Understanding Data

Subtopic: Types of Data — Structured vs. Unstructured

1. Data organised in rows and columns is called:
- A) Unstructured data
 - B) Structured data**
 - C) Raw data
 - D) Random data
2. Which of the following is the BEST example of structured data?
- A) A voice message recording
 - B) A student mark sheet stored in a spreadsheet**
 - C) A photograph of a classroom
 - D) A social media post with emojis
3. Which of the following is an example of unstructured data?
- A) A database of bank transactions
 - B) A spreadsheet of product prices
 - C) A video recording of a lecture**
 - D) A table of student attendance percentages

4. Why is unstructured data considered harder to analyse than structured data?
- A) It contains fewer data points
 - B) It does not follow a fixed format, so special tools are required to process it**
 - C) It is always encrypted
 - D) It can only be stored on cloud systems
5. Emails, images, audio recordings, and videos are all examples of:
- A) Structured data
 - B) Unstructured data**
 - C) Relational data
 - D) Tabular data
6. Approximately what percentage of the world's total data is unstructured?
- A) 20%
 - B) 40%
 - C) 80%**
 - D) 60%
7. A hospital database storing patient names, ages, diagnoses, and test results in a table is an example of:
- A) Structured data**
 - B) Unstructured data
 - C) Audio data
 - D) Spatial data
8. What tool is typically needed to process unstructured data like voice recordings?
- A) A simple spreadsheet formula
 - B) Specialised tools such as speech-to-text or natural language processing**
 - C) A standard database query
 - D) A basic sorting algorithm
9. Social media posts, tweets, and blog articles are best classified as:
- A) Structured data
 - B) Unstructured data**
 - C) Relational data
 - D) Numerical data
10. Which of the following pairs correctly matches the data type to its example?
- A) Structured data → Audio file; Unstructured data → Spreadsheet
 - B) Structured data → Database table; Unstructured data → Video file**
 - C) Structured data → Email text; Unstructured data → Excel table
 - D) Structured data → Image; Unstructured data → CSV file

Subtopic: Data Sources and Data Collection Methods

1. Which of the following is an example of AUTOMATIC data collection?
 - A) A researcher manually recording observations in a notebook
 - B) A temperature sensor recording readings every minute**
 - C) A teacher entering student grades into a spreadsheet by hand
 - D) A student filling out a paper survey
2. Which of the following is a common SOURCE of data?
 - A) Computer hardware chips
 - B) Online platforms, sensors, surveys, and interviews**
 - C) Programming languages
 - D) Operating systems
3. Manual data collection differs from automatic data collection because:
 - A) Manual collection uses sensors while automatic uses humans
 - B) Manual collection requires human effort while automatic uses machines or software**
 - C) Manual collection is always faster
 - D) Automatic collection only works offline
4. Real-time data collection means:
 - A) Data is saved and analysed at the end of the month
 - B) Data is captured and processed immediately as events happen**
 - C) Data is collected once per year
 - D) Data is only collected from websites
5. Which of the following describes "batch data collection"?
 - A) Data is processed the moment it is generated
 - B) Data is collected using sensors only
 - C) Data is stored over time and analysed later as a group**
 - D) Data is collected from a single source automatically
6. The phrase "Garbage In, Garbage Out" in data science refers to:
 - A) Physical waste in server rooms
 - B) The idea that poor quality data leads to incorrect and unreliable analysis results**
 - C) Spam emails corrupting databases
 - D) Errors in hardware manufacturing
7. Which of the following collection methods is most likely to capture data from thousands of respondents quickly?
 - A) Personal interviews
 - B) Manual observation
 - C) Online surveys and forms**
 - D) Physical paper forms only

Subtopic: Data Storage and Management Concepts

1. Which of the following is the best definition of "data storage"?
 - A) Analysing data for patterns
 - B) Saving data in files, databases, or cloud systems for future use**
 - C) Transmitting data across the internet
 - D) Displaying data in charts and graphs
2. Which of the following is NOT a type of data storage system?
 - A) Relational database
 - B) Cloud storage
 - C) A programming compiler**
 - D) Data warehouse
3. The "3-2-1 backup rule" in data management means:
 - A) Back up data once a year on 3 devices
 - B) Keep 3 copies of data, on 2 different media types, with 1 copy stored offsite**
 - C) Delete old data every 3 months from 2 locations
 - D) Use 1 cloud server with 2 local backups deleted after 3 months
4. "Data integrity" refers to:
 - A) The speed at which data is transferred
 - B) Ensuring data remains accurate, consistent, and uncorrupted over time**
 - C) The physical security of server rooms
 - D) The number of users who can access data simultaneously
5. Which of the following is a common problem found in real-world datasets that requires cleaning?
 - A) Data that is perfectly formatted
 - B) Missing values, duplicate records, and incorrect data types**
 - C) Data stored in cloud systems
 - D) Data collected from sensors
6. A data lake is best described as:
 - A) A physical location near water for server placement
 - B) A storage system that holds raw data in any format until it is needed for analysis**
 - C) A type of structured database using only tables
 - D) A backup drive for a single user
7. Which technique handles missing values by replacing them with the average value of that column?
 - A) One-hot encoding
 - B) Mean imputation**
 - C) Feature scaling
 - D) Data splitting

8. Approximately how much of a data scientist's project time is spent on data cleaning?
- A) 5–10%
 - B) 20–30%
 - C) 60–80%**
 - D) 90–95%
9. Which of the following is an example of a data management concern?
- A) Choosing the colour of a website
 - B) Protecting data from unauthorised access using passwords and encryption**
 - C) Selecting a programming language for web development
 - D) Writing user interface code
10. An outlier in a dataset is:
- A) A correctly formatted data value
 - B) A data value that is abnormally different from the rest of the dataset**
 - C) A missing value in a table
 - D) A duplicate record in a database
-

Topic 6.3: Overview of Machine Learning

1. Machine learning is best described as:
- A) A method for designing computer hardware
 - B) A technique that enables systems to learn from data and improve without being explicitly programmed**
 - C) A type of database management system
 - D) A programming language used to build websites
2. In traditional programming, the output is produced by:
- A) Analysing patterns in large datasets
 - B) Following fixed rules written by a programmer**
 - C) Rewarding correct decisions
 - D) Grouping data into clusters
3. In machine learning, rules are:
- A) Written manually by programmers
 - B) Hard-coded into the system
 - C) Discovered automatically by the system from data**
 - D) Retrieved from the internet

4. Which of the following is the best analogy for machine learning?
- A) A chef following a strict recipe step by step
 - B) A chef studying 1,000 photos of dishes and figuring out the recipe themselves**
 - C) A chef reading a menu to a customer
 - D) A chef ordering ingredients from a supplier
5. In traditional programming, what happens when the rules need to change?
- A) The data is automatically updated
 - B) A programmer must write new code**
 - C) The machine learning model retrains itself
 - D) The database is replaced
6. Machine learning differs from traditional programming because:
- A) Machine learning requires more programmers
 - B) Machine learning uses data to learn rules automatically instead of following hard-coded rules**
 - C) Traditional programming can handle more complex problems
 - D) Machine learning only works for image data
7. Which of the following problems is BEST suited for machine learning rather than traditional programming?
- A) Calculating a student's average mark using a formula
 - B) Recognising a person's face in a photograph under varying lighting conditions**
 - C) Sorting a list of names in alphabetical order
 - D) Converting temperatures from Celsius to Fahrenheit
8. "Learning from data" in machine learning means:
- A) Downloading textbooks from the internet
 - B) Using past information and examples to improve future predictions and decisions**
 - C) Reading data stored in a database without analysing it
 - D) Copying data from one system to another
9. Machine learning is described as a subfield of:
- A) Hardware engineering
 - B) Network administration
 - C) Data science and artificial intelligence**
 - D) Web development
10. As a machine learning model processes more data, it typically becomes:
- A) Slower and less efficient
 - B) Unchanged in its performance
 - C) More accurate and efficient**
 - D) Less relevant to new problems
-

Topic 6.4: Machine Learning Mechanisms

Subtopic: Supervised Machine Learning

1. Supervised machine learning uses:
 - A) Unlabelled data and finds its own patterns
 - B) Labelled data where each input has a known correct output**
 - C) Rewards and penalties to guide learning
 - D) Random data without any structure
2. In supervised learning, the model learns by:
 - A) Grouping similar data items together
 - B) Comparing its predictions to correct answers and reducing the error**
 - C) Interacting with an environment and receiving rewards
 - D) Randomly guessing until it is correct
3. Which of the following is an example of labelled data?
 - A) A collection of photographs with no captions
 - B) Emails marked as "spam" or "not spam" by a human reviewer**
 - C) Customer transaction records without categories
 - D) Audio files with no transcriptions
4. Supervised learning is particularly useful for which type of task?
 - A) Discovering hidden clusters in data
 - B) Training an agent to play a video game
 - C) Predicting a known output type from input features**
 - D) Reducing the size of a dataset
5. Which of the following is NOT an application of supervised learning?
 - A) Email spam detection
 - B) Medical diagnosis from scans
 - C) Grouping customers by behaviour without predefined categories**
 - D) Predicting student exam results
6. A task where the model predicts a CATEGORY (e.g., "cat" or "dog") is called:
 - A) Regression
 - B) Classification**
 - C) Clustering
 - D) Reinforcement

7. A task where the model predicts a CONTINUOUS NUMBER (e.g., house price) is called:
- A) Regression**
 - B) Classification
 - C) Clustering
 - D) Encoding
8. Face recognition on a smartphone is an example of:
- A) Supervised learning (classification)**
 - B) Unsupervised learning (clustering)
 - C) Reinforcement learning
 - D) Traditional rule-based programming
9. Model accuracy in supervised learning generally improves when:
- A) The model is made more complex regardless of data size
 - B) The model is trained on more high-quality labelled examples**
 - C) The validation set is included in the training data
 - D) Features are removed from the dataset
10. Which of the following is an example of a REGRESSION task in supervised learning?
- A) Classifying emails as spam or not spam
 - B) Detecting whether a medical scan shows cancer or not
 - C) Predicting the price of a house based on its features**
 - D) Identifying whether a photo contains a cat or a dog

Subtopic: Unsupervised Machine Learning

1. Unsupervised machine learning works with:
- A) Labelled data with correct answers provided
 - B) Unlabelled data where the model finds patterns on its own**
 - C) Data collected only from sensors
 - D) Structured tabular data only
2. The most common technique in unsupervised learning is:
- A) Regression
 - B) Classification
 - C) Clustering**
 - D) Reinforcement
3. What does clustering do in unsupervised learning?
- A) Assigns every data point to a pre-defined category
 - B) Groups similar data points together without being told what the groups should be called**
 - C) Predicts a numerical value for each data point
 - D) Removes outliers from the dataset automatically

4. Which of the following is an example of unsupervised learning?
- A) Training a spam filter using labelled emails
 - B) Grouping customers into segments based on purchasing behaviour without predefined groups**
 - C) Predicting exam scores from study hours
 - D) Teaching a robot to navigate by rewarding correct moves
5. In unsupervised learning, no correct output is provided because:
- A) The dataset is too small
 - B) The goal is to discover hidden patterns and structures that are not known in advance**
 - C) The model is too complex
 - D) The data has already been labelled
6. Anomaly detection — identifying unusual patterns that do not match normal behaviour — is an application of:
- A) Unsupervised learning**
 - B) Supervised classification
 - C) Reinforcement learning
 - D) Linear regression
7. A news website uses a system that automatically groups thousands of articles by topic without human labelling. This is an example of:
- A) Supervised classification
 - B) Unsupervised clustering**
 - C) Reinforcement learning
 - D) Feature engineering

Subtopic: Reinforcement Machine Learning

1. In reinforcement machine learning, what guides the learning process?
- A) Labelled datasets with correct outputs
 - B) Fixed rules written by programmers
 - C) Rewards for correct actions and penalties for incorrect actions**
 - D) Predefined clusters in the data
2. The "agent" in reinforcement learning refers to:
- A) The dataset used for training
 - B) The model or system that takes actions in an environment**
 - C) The reward signal
 - D) The programmer who designs the system

3. In reinforcement learning, a "positive reward" encourages the agent to:
- A) Stop and restart training from scratch
 - B) Repeat and reinforce the behaviour that earned the reward**
 - C) Choose a different algorithm
 - D) Reduce the size of the training dataset
4. Which of the following is the best analogy for reinforcement learning?
- A) Studying flashcards with answers on the back
 - B) Sorting a pile of Lego bricks by shape with no instructions
 - C) Training a dog with treats for good behaviour and "no" for bad behaviour**
 - D) Filling in a multiple-choice test with 50/50 guessing
5. AlphaGo, the AI that mastered the board game Go, is an example of:
- A) Supervised learning
 - B) Unsupervised clustering
 - C) Reinforcement learning**
 - D) Traditional rule-based programming
6. In reinforcement learning, the term "environment" refers to:
- A) The size of the training dataset
 - B) The external world in which the agent takes actions and receives feedback**
 - C) The validation set used for tuning
 - D) The list of hyperparameters
7. A self-driving car that learns to make driving decisions by simulating millions of driving scenarios with rewards for safe driving and penalties for crashes is using:
- A) Supervised learning
 - B) Unsupervised learning
 - C) Reinforcement learning**
 - D) Feature engineering
8. The goal of the agent in reinforcement learning is to:
- A) Memorise the training dataset
 - B) Maximise total accumulated rewards over time**
 - C) Minimise the number of features used
 - D) Reduce the size of the dataset
9. Which of the following correctly matches all three ML types to their learning style?
- A) Supervised=rewards, Unsupervised=labels, Reinforcement=clusters
 - B) Supervised=labelled examples, Unsupervised=no labels, Reinforcement=trial and error**
 - C) Supervised=clusters, Unsupervised=rewards, Reinforcement=labels
 - D) Supervised=no labels, Unsupervised=rewards, Reinforcement=labelled examples
-

Topic 6.5: Applications of Machine Learning

1. In healthcare, machine learning helps detect cancerous cells by analysing:
 - A) Patient billing records only
 - B) Medical images such as MRI scans, X-rays, and CT scans**
 - C) Appointment scheduling calendars
 - D) Pharmacy inventory databases
2. Which machine learning application helps banks identify suspicious financial transactions in real time?
 - A) Customer segmentation
 - B) Fraud detection**
 - C) Inventory management
 - D) Sales forecasting
3. A recommendation system that suggests movies based on what a user previously watched uses:
 - A) Unsupervised clustering only
 - B) Machine learning to predict what content the user is most likely to enjoy next**
 - C) Traditional rule-based filtering only
 - D) Manual human curation
4. The Netflix Prize (2006–2009) was awarded for:
 - A) Designing the Netflix streaming infrastructure
 - B) Improving Netflix's movie recommendation algorithm by at least 10%**
 - C) Building the first machine learning model
 - D) Developing the Python programming language
5. Machine learning in education can help by:
 - A) Replacing all teachers with robots
 - B) Identifying students at risk of falling behind and personalising learning content**
 - C) Only grading multiple-choice tests
 - D) Managing school building maintenance schedules
6. Which of the following is an everyday application of machine learning that most people use daily?
 - A) Manually sorting email into folders
 - B) Spam filters that automatically classify unwanted emails**
 - C) Writing code in a text editor
 - D) Using a calculator for arithmetic

7. Google Maps suggesting the fastest route based on current traffic conditions uses:
- A) Traditional rule-based navigation only
 - B) Machine learning models that predict real-time traffic patterns**
 - C) Satellite imagery without any data analysis
 - D) Manual input from human traffic controllers
8. Personalised learning in education — where content difficulty adapts to each student's progress — is an application of:
- A) Machine learning**
 - B) Traditional database management
 - C) Hardware engineering
 - D) Network administration
9. Which of the following is NOT a healthcare application of machine learning?
- A) Early detection of Alzheimer's disease
 - B) Assisting in cardiovascular disease diagnosis
 - C) Managing office supply inventory for a hospital**
 - D) Identifying cancer in medical scan images
10. The use of chatbots in customer service is an application of machine learning in:
- A) Healthcare
 - B) Business**
 - C) Education
 - D) Transportation
11. Smart home devices that learn a user's temperature preferences and automate the thermostat are an example of:
- A) Supervised learning in healthcare
 - B) Everyday machine learning applications**
 - C) Unsupervised clustering in business
 - D) Reinforcement learning in robotics only
12. Drug discovery using machine learning speeds up the process by:
- A) Replacing laboratory scientists entirely
 - B) Simulating how molecules interact with biological systems to identify promising compounds faster**
 - C) Managing hospital appointment systems
 - D) Automating patient billing
-

Topic 6.6: Building a Machine Learning Model

Subtopic: Selecting Features and Target Values

1. In a machine learning model, "features" are:
 - A) The output values the model predicts
 - B) The input columns that provide information the model uses to make predictions**
 - C) The errors produced during training
 - D) The hyperparameter values chosen before training
2. The "target value" in a machine learning model is:
 - A) The first feature column in the dataset
 - B) The output value the model is trained to predict**
 - C) The number of rows in the training dataset
 - D) The validation score of the model
3. In a student grade prediction model, which of the following would be a FEATURE?
 - A) Final exam grade (what we want to predict)
 - B) Study hours per day**
 - C) Student ID number
 - D) The model's accuracy score
4. Which column should be REMOVED during feature selection for a student grade prediction model?
 - A) Attendance percentage
 - B) Student ID number**
 - C) Quiz average score
 - D) Study hours per week
5. Why is proper feature selection important?
 - A) It increases the amount of data available
 - B) It removes irrelevant information that adds noise and reduces model accuracy**
 - C) It automatically increases model complexity
 - D) It eliminates the need for a train-test split
6. In a house price prediction model, which of the following is the TARGET value?
 - A) Number of bedrooms
 - B) Neighbourhood
 - C) Sale price of the house**
 - D) Year the house was built

Subtopic: Feature Engineering

1. Feature engineering is best defined as:
 - A) Writing new machine learning algorithms from scratch
 - B) Transforming and preparing raw data so that a model can learn from it more effectively**
 - C) Collecting new data from external sources
 - D) Choosing which programming language to use
2. One-Hot Encoding is used in feature engineering to:
 - A) Scale numerical features to the range 0 to 1
 - B) Convert text categories into binary (0 or 1) columns so models can use them**
 - C) Fill in missing values with the column mean
 - D) Remove duplicate records from a dataset
3. A dataset has a column "City" with values: Karachi, Lahore, Islamabad. After One-Hot Encoding, how many new columns are created?
 - A) 1
 - B) 2
 - C) 3**
 - D) 4
4. Feature scaling is important because:
 - A) It reduces the number of rows in the training set
 - B) It ensures all features are on a similar numerical scale so no single feature dominates the model unfairly**
 - C) It removes all categorical columns from the dataset
 - D) It automatically selects the best features for the model
5. Min-Max Scaling transforms a feature value using the formula:
 - A) $(\text{value} + \text{mean}) / \text{standard deviation}$
 - B) $(\text{value} - \text{minimum}) / (\text{maximum} - \text{minimum})$**
 - C) $\text{value} \times (\text{maximum} - \text{minimum})$
 - D) $(\text{value} - \text{maximum}) / (\text{minimum} + \text{maximum})$
6. After applying Min-Max Scaling, all feature values fall within the range:
 - A) -1 to +1
 - B) 0 to 1**
 - C) 0 to 100
 - D) $-\infty$ to $+\infty$
7. Creating a new feature "Study_Hours_Per_Day" from "Total_Weekly_Study_Hours / 5" is an example of:
 - A) One-Hot Encoding

- B) Mean imputation
- C) Feature creation (a form of feature engineering)**
- D) Train-test splitting

8. Which of the following correctly describes mean imputation for missing values?
- A) Deleting all rows that contain any missing value
 - B) Replacing missing values in a column with the average value of that column**
 - C) Replacing missing values with zero
 - D) Flagging missing values with a special character

Subtopic: Train-Test Split

1. The purpose of splitting data into a training set and a test set is to:
 - A) Make the dataset smaller and easier to manage
 - B) Evaluate how well the model performs on new, unseen data it was not trained on**
 - C) Increase the number of features available to the model
 - D) Remove outliers from the dataset
2. What percentage of data is typically used for the TRAINING set in an 80/20 split?
 - A) 80%**
 - B) 20%
 - C) 50%
 - D) 70%
3. Why must the model NEVER see the test set during training?
 - A) The test set contains incorrect data
 - B) Seeing the test set would allow the model to memorise answers rather than truly learn patterns**
 - C) The test set is stored in a different format
 - D) Training on test data makes the model faster
4. "Overfitting" in machine learning means:
 - A) The model is too simple to capture any patterns
 - B) The model memorises training data and performs poorly on new, unseen data**
 - C) The model was trained on too little data
 - D) The model was trained with too many features removed
5. "Underfitting" in machine learning means:
 - A) The model memorises training data perfectly
 - B) The model is too simple and fails to capture the patterns even in the training data**
 - C) The model was trained on too many features
 - D) The model's accuracy is above 90%

6. In the study guide's exam paper analogy, the training set corresponds to:
- A) The brand-new exam questions on exam day
 - B) Past exam papers used for studying and practising**
 - C) The teacher who grades the exam
 - D) The classroom where the exam is held
7. The `random_state=42` parameter in scikit-learn's `train_test_split` function ensures:
- A) 42% of data goes to the test set
 - B) The same random split is reproduced every time the code is run**
 - C) The model uses 42 features for training
 - D) The training process runs for exactly 42 iterations
8. For a dataset with millions of records, a common training/test split might be:
- A) 50% / 50%
 - B) 80% / 20%
 - C) 99% / 1% (since 1% of millions is still thousands of test examples)**
 - D) 30% / 70%

Subtopic: Building a Simple Predictive Model

1. Linear regression is a machine learning algorithm that:
- A) Groups data into clusters
 - B) Finds the best straight line that models the relationship between input features and a numerical output**
 - C) Assigns data to categories using labels
 - D) Maximises rewards through trial and error
2. In the linear regression model equation `Exam Score = 7.55 × Hours + 16.9`, the value 7.55 represents:
- A) The minimum possible exam score
 - B) The increase in predicted exam score for every additional hour studied**
 - C) The total number of hours studied
 - D) The intercept of the regression line
3. In the same equation, 16.9 (the intercept) represents:
- A) The slope of the regression line
 - B) The predicted exam score when a student studies 0 hours**
 - C) The maximum possible exam score
 - D) The average number of hours studied

4. Mean Squared Error (MSE) is used to:
- A) Measure how many predictions were correct
 - B) Measure the average squared difference between predicted values and actual values**
 - C) Calculate the F1-score of a classifier
 - D) Determine the number of features in the model
5. A lower MSE value indicates:
- A) More errors in the model's predictions
 - B) Predicted values are closer to the actual values — better model performance**
 - C) The model is overfitting the training data
 - D) The model needs more features
6. The R-squared (R^2) metric in regression tells us:
- A) How many rows are in the training set
 - B) What proportion of the variation in the target variable is explained by the model**
 - C) The number of incorrect predictions
 - D) The model's precision score
7. An R^2 value of 0.97 means:
- A) The model made 97 errors
 - B) The model is 97% overfit
 - C) The model explains 97% of the variation in the target variable — an excellent fit**
 - D) The training set contains 97% of the data
-

Topic 6.7: Model Evaluation and Performance Metrics

Subtopic: The Confusion Matrix

1. A confusion matrix is used to:
- A) Show the hyperparameters used to train a model
 - B) Break down all predictions into TP, TN, FP, and FN categories for evaluation**
 - C) Display the architecture of a neural network
 - D) Compare the sizes of the training and test sets
2. "True Positive (TP)" in a confusion matrix means:
- A) The model predicted negative and was correct
 - B) The model predicted positive and the actual value was also positive**
 - C) The model predicted negative but the actual value was positive
 - D) The model predicted positive but the actual value was negative

3. "False Positive (FP)" is also known as a:
- A) Type I Error — the model incorrectly predicted positive when the actual value was negative**
 - B) Type II Error — the model missed a real positive case
 - C) A correct prediction of a positive case
 - D) A correct prediction of a negative case
4. "False Negative (FN)" is also known as a:
- A) Type I Error — the model incorrectly predicted positive
 - B) Type II Error — the model predicted negative but the actual value was positive**
 - C) A correct prediction of a positive case
 - D) A correct prediction of a negative case
5. "True Negative (TN)" in a confusion matrix means:
- A) The model predicted positive and was correct
 - B) The model predicted positive but was wrong
 - C) The model predicted negative and the actual value was also negative — a correct prediction**
 - D) The model predicted negative but the actual value was positive

Subtopic: Accuracy and Error Rate

1. The formula for Accuracy is:
- A) $TP / (TP + FP)$
 - B) $TP / (TP + FN)$
 - C) $(TP + TN) / (TP + TN + FP + FN)$**
 - D) $2 \times (Precision \times Recall) / (Precision + Recall)$
2. A model achieves 99% accuracy on a medical dataset where 99% of patients are healthy. This result is:
- A) Excellent — the model should be deployed immediately
 - B) Misleading — the model likely predicts "healthy" for all patients and misses all sick cases**
 - C) A sign of underfitting
 - D) Impossible in practice
3. Accuracy is most reliable as a metric when:
- A) The dataset has many missing values
 - B) The dataset is balanced, with roughly equal numbers of examples from each class**
 - C) The model is overfitting
 - D) The dataset is highly imbalanced

4. If TP=40, TN=50, FP=10, FN=5, what is the Accuracy?
- A) 80.0%
 - B) 85.7%**
 - C) 88.9%
 - D) 84.2%
5. Error Rate is calculated as:
- A) $TP / (TP + FP)$
 - B) $1 - \text{Accuracy}$ (or $FP + FN$ divided by total predictions)**
 - C) $(TP + TN) / \text{Total}$
 - D) $TP / (TP + FN)$

Subtopic: Precision and Recall

1. Precision measures:
- A) How many actual positive cases the model correctly identified
 - B) Of all cases the model predicted as positive, what fraction were actually positive**
 - C) The overall fraction of correct predictions
 - D) The balance between false positives and false negatives
2. The formula for Precision is:
- A) $TP / (TP + FP)$**
 - B) $TP / (TP + FN)$
 - C) $(TP + TN) / (TP + TN + FP + FN)$
 - D) $2 \times (P \times R) / (P + R)$
3. Recall measures:
- A) How many of the model's positive predictions were correct
 - B) Of all actual positive cases, what fraction did the model correctly identify**
 - C) The model's overall accuracy on the test set
 - D) The ratio of true negatives to total predictions
4. The formula for Recall is:
- A) $TP / (TP + FP)$
 - B) $TP / (TP + FN)$**
 - C) $(TP + TN) / \text{Total}$
 - D) $2 \times P \times R / (P + R)$
5. If TP=40, FP=10, what is the Precision?
- A) 0.80 (80%)**
 - B) 0.89 (89%)
 - C) 0.84 (84%)
 - D) 0.57 (57%)

6. If TP=40, FN=5, what is the Recall?
- A) 0.80 (80%)
 - B) 0.889 (88.9%)**
 - C) 0.842 (84.2%)
 - D) 0.571 (57.1%)
7. In a cancer detection system, which metric is most critical to maximise and why?
- A) Precision — because wrongly diagnosing a healthy patient as sick is catastrophic
 - B) Recall — because missing a real cancer case (False Negative) could be fatal**
 - C) Accuracy — because the overall correctness rate is most important
 - D) Error Rate — because we want to minimise all types of errors equally
8. In an email spam filter, which metric is most critical to maximise?
- A) Precision — because sending a real, important email to the spam folder (False Positive) is very costly**
 - B) Recall — because letting some spam through is the biggest concern
 - C) Accuracy — because overall correctness is all that matters
 - D) F1-Score is irrelevant for this type of problem
9. The Precision-Recall trade-off means:
- A) Increasing both precision and recall simultaneously is always easy
 - B) Attempting to maximise one often causes the other to decrease**
 - C) Precision and recall always have the same value
 - D) Only one of these metrics can be calculated at a time

Subtopic: F1-Score

1. The F1-Score is defined as:
- A) $(TP + TN) / \text{Total}$
 - B) $TP / (TP + FP)$
 - C) $2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$**
 - D) $TP / (TP + FN)$
2. The F1-Score is called the "harmonic mean" of precision and recall because:
- A) It is calculated by adding precision and recall and dividing by 2
 - B) It gives a lower result when either precision or recall is very low, punishing extreme imbalances**
 - C) It multiplies precision by recall directly
 - D) It only applies to balanced datasets

3. If Precision = 0.80 and Recall = 0.8889, what is the F1-Score?
- A) 0.8445
 - B) 0.8421**
 - C) 0.8000
 - D) 0.8889
4. F1-Score is most useful when:
- A) The dataset is perfectly balanced between classes
 - B) You only care about precision
 - C) The dataset is imbalanced and you need a metric that balances both precision and recall**
 - D) You are building a regression model, not a classifier
5. A model with very high Recall but very low Precision would have:
- A) A high F1-Score, because recall dominates
 - B) A low F1-Score, because the harmonic mean is dragged down by the low precision**
 - C) An F1-Score equal to the recall value
 - D) An F1-Score that cannot be calculated
6. Which metric should be used when a data scientist needs ONE number to summarise classifier performance on an imbalanced dataset?
- A) Accuracy
 - B) Precision alone
 - C) Recall alone
 - D) F1-Score**
-

Topic 6.8: Validation and Model Improvement

Subtopic: Validation Dataset Preparation

1. A validation dataset is used for:
- A) Training the model and learning patterns from data
 - B) Tuning hyperparameters and comparing model configurations without contaminating the test set**
 - C) The final unbiased evaluation of model performance
 - D) Storing data that is not yet cleaned
2. In a three-way data split, the CORRECT order of usage is:
- A) Test → Validation → Training
 - B) Training (learn) → Validation (tune) → Test (final evaluation)**
 - C) Validation → Training → Test
 - D) Training → Test → Validation

3. Why should hyperparameters NEVER be tuned using the test set?
- A) The test set is too small for tuning
 - B) Using the test set for tuning means the model indirectly "sees" test data, making the final evaluation biased**
 - C) The test set does not contain labelled data
 - D) Tuning on the test set causes underfitting
4. In K-Fold Cross Validation with K=5, how many times is the model trained and validated?
- A) 1
 - B) 3
 - C) 5**
 - D) 10
5. K-Fold Cross Validation is especially useful when:
- A) The dataset is extremely large and needs to be reduced
 - B) The dataset is limited in size and a single validation split might give unreliable estimates**
 - C) The model has already been deployed to production
 - D) The training process is too fast
6. A common three-way data split ratio for medium-sized datasets is:
- A) 90% train / 5% validation / 5% test
 - B) 70% train / 15% validation / 15% test**
 - C) 50% train / 25% validation / 25% test
 - D) 80% train / 10% validation / 10% test

Subtopic: Model Robustness and Reliability

1. A "robust" machine learning model is one that:
- A) Achieves 100% accuracy on the training set
 - B) Performs consistently well across different data conditions, including noisy or slightly different data**
 - C) Uses the most complex algorithm available
 - D) Requires the least amount of training data
2. A model that scores 99% accuracy on training data but only 65% on test data is most likely:
- A) Underfitting
 - B) Overfitting**
 - C) Properly tuned
 - D) Robust and ready for deployment

3. Which of the following is NOT a strategy for building a more robust model?
- A) Collecting more diverse training data
 - B) Using regularisation to penalise overly complex models
 - C) Including the test set data in the training process to increase data volume**
 - D) Using K-Fold cross-validation
4. Reliability in a machine learning model means:
- A) The model always outputs the same answer regardless of input
 - B) The model gives consistent, trustworthy predictions when applied to new real-world data over time**
 - C) The model was built using the most popular algorithm
 - D) The model's code has no syntax errors

Subtopic: Hyperparameter Tuning

1. Hyperparameters differ from model parameters because:
- A) Hyperparameters are learned from the data during training; parameters are set before training
 - B) Hyperparameters are set by the engineer before training; parameters are learned from the data**
 - C) Both are learned from the data during training
 - D) Both are set manually and never change
2. Which of the following is an example of a hyperparameter?
- A) The slope (coefficient) in a linear regression model
 - B) The weights in a neural network
 - C) The maximum depth allowed for a decision tree (max_depth)**
 - D) The predicted output value for a new input
3. The "speaker volume" analogy for hyperparameter tuning illustrates that:
- A) Louder models always perform better
 - B) Settings that are too high (overfitting) or too low (underfitting) both cause problems — the goal is the right balance**
 - C) Hyperparameter tuning is only relevant for audio processing models
 - D) All models should use the same hyperparameter values
4. The process of testing different hyperparameter values and selecting the one with the best validation performance is called:
- A) Feature engineering
 - B) Data cleaning
 - C) Hyperparameter tuning**
 - D) Train-test splitting

5. In a K-Nearest Neighbours (KNN) model, what happens when K is set to 1?
- A) The model becomes very generalised and ignores most data points
 - B) The model becomes very sensitive to individual data points, likely overfitting to the training data**
 - C) The model ignores all features
 - D) The model always predicts the majority class
6. Which dataset is used to select the best hyperparameter values?
- A) The training set
 - B) The validation set**
 - C) The test set
 - D) The full dataset before splitting
-

Topic 6.9: Predictive Modeling and Causality

1. Predictive modelling in machine learning is used to:
- A) Explain why past events happened
 - B) Estimate what will happen in the future based on patterns found in historical data**
 - C) Store data in an organised format
 - D) Clean and transform raw datasets
2. Which of the following is an example of a predictive outcome in machine learning?
- A) Discovering that a water pump caused a cholera outbreak
 - B) Estimating tomorrow's weather based on today's temperature and humidity patterns**
 - C) Proving that smoking directly causes lung cancer
 - D) Running an experiment to test whether tutoring improves grades
3. Causality in data science refers to:
- A) The ability of a model to make accurate predictions
 - B) A direct cause-and-effect relationship where one factor directly influences another**
 - C) The correlation between two variables that move together
 - D) The process of training a model on labelled data
4. Which of the following is the best example of a causal relationship?
- A) Ice cream sales and drowning rates both increase in summer
 - B) Students with water bottles score higher on exams
 - C) Regular exercise directly lowers blood pressure through cardiovascular changes**
 - D) Rooster crowing and sunrise occurring at the same time
5. Two variables are "correlated" when:
- A) One definitively causes a change in the other
 - B) They tend to move together (both increase or both decrease) without necessarily**

having a causal link

- C) They are measured using the same units
 - D) They are both categorical variables
6. The "ice cream and drowning" example demonstrates:
- A) That ice cream is dangerous and causes accidents
 - B) That correlation does not equal causation — a hidden third variable (summer heat) causes both**
 - C) That machine learning models cannot distinguish correlation from causation
 - D) That predictive models are always wrong
7. To prove causation scientifically, the most reliable method is:
- A) Finding a high correlation coefficient between two variables
 - B) Building a machine learning model that predicts one variable from another
 - C) Conducting a controlled experiment (Randomised Control Trial) that isolates the causal variable**
 - D) Collecting more data until the pattern becomes clearer
8. A machine learning model can make accurate predictions:
- A) Only when it has identified the true causal variables
 - B) Even when the patterns it uses are correlational rather than causal**
 - C) Only when the dataset contains experimental data
 - D) Only when all features have causal relationships with the target
9. The difference between prediction and causation is best summarised as:
- A) Prediction explains why; causation forecasts what will happen
 - B) Prediction forecasts what will happen; causation explains why it happens**
 - C) They are identical concepts in data science
 - D) Causation is a type of supervised learning; prediction is a type of unsupervised learning
10. In the student performance example from the chapter, study hours predicting exam scores is:
- A) Pure causation with no confounding variables
 - B) A predictive relationship — the model forecasts scores, but motivation may be the real causal driver**
 - C) An example of unsupervised learning
 - D) An example of reinforcement learning
-

Topic 6.10: Machine Learning Tools and Platforms

Subtopic: Excel

1. Microsoft Excel is best suited for:
 - A) Training deep learning models on millions of records
 - B) Organising, visualising, and performing basic calculations on small to medium datasets**
 - C) Building and deploying production machine learning systems
 - D) Web scraping large amounts of data from the internet
2. Which of the following is a feature of Excel for data analysis?
 - A) Running Python machine learning code
 - B) Creating bar charts, line graphs, and scatter plots from data**
 - C) Building neural networks
 - D) Performing automatic feature engineering
3. What is the primary limitation of Excel for data science?
 - A) It cannot create any visualisations
 - B) It struggles with very large datasets and lacks built-in machine learning capabilities**
 - C) It requires advanced programming knowledge to use
 - D) It cannot perform any calculations
4. An Excel formula to calculate the average of values in cells B2 to B100 is:
 - A) =SUM(B2:B100)
 - B) =AVERAGE(B2:B100)**
 - C) =MEAN(B2:B100)
 - D) =CALC(B2:B100)

Subtopic: R for Data Science

1. The R programming language was designed primarily for:
 - A) Building websites and web applications
 - B) Statistical analysis and data visualisation**
 - C) Network administration and cybersecurity
 - D) Operating system development
2. Which R library is widely used for creating high-quality data visualisations?
 - A) scikit-learn
 - B) pandas
 - C) ggplot2**
 - D) NumPy

3. R is most commonly used in which context?
- A) Mobile app development
 - B) Academic research and scientific data analysis**
 - C) Video game development
 - D) Operating system programming
4. The R command `summary(dataset)` produces:
- A) A bar chart of all columns
 - B) Basic statistical summaries (mean, median, min, max, quartiles) for all columns**
 - C) A sorted list of all values
 - D) A confusion matrix
5. In R, the function `lm(y ~ x, data = dataset)` is used to:
- A) Create a bar chart of column x
 - B) Fit a linear regression model predicting y from x**
 - C) Calculate the F1-Score
 - D) Perform K-Fold cross-validation

Subtopic: Python and Jupyter Notebooks

1. Python is described as the most popular language for machine learning because:
- A) It is the fastest programming language available
 - B) It has a rich ecosystem of powerful, easy-to-use libraries like pandas, NumPy, and scikit-learn**
 - C) It was designed exclusively for data science
 - D) It does not require any installation
2. Which Python library is used for working with data tables (similar to Excel in code)?
- A) pandas**
 - B) NumPy
 - C) matplotlib
 - D) scikit-learn
3. Which Python library provides ready-to-use machine learning algorithms including linear regression, KNN, and decision trees?
- A) pandas
 - B) NumPy
 - C) scikit-learn**
 - D) matplotlib

4. Which Python library is used for creating charts and plots?
- A) pandas
 - B) scikit-learn
 - C) matplotlib**
 - D) NumPy
5. NumPy in Python is primarily used for:
- A) Creating web applications
 - B) Fast numerical calculations and working with arrays of numbers**
 - C) Building neural networks only
 - D) Managing database connections
6. Jupyter Notebook is best described as:
- A) A Python library for machine learning
 - B) An interactive coding environment where code, output, and explanatory text can be combined in one document**
 - C) A cloud storage system for datasets
 - D) A type of database for structured data
7. In Jupyter Notebook, code is written and executed in:
- A) A terminal command line only
 - B) Individual cells that can be run independently**
 - C) A single file that must be compiled first
 - D) A web browser's address bar
8. The Python function `train_test_split(X, y, test_size=0.2, random_state=42)` splits data so that:
- A) 20% goes to training and 80% goes to testing
 - B) 80% goes to training and 20% goes to testing**
 - C) 42% goes to training and 58% goes to testing
 - D) All data goes to training with 20% reserved for validation
9. The scikit-learn function `model.fit(X_train, y_train)` does which of the following?
- A) Tests the model on unseen data and returns accuracy
 - B) Trains the model by finding patterns in the training features and labels**
 - C) Creates the confusion matrix for evaluation
 - D) Splits the data into training and test sets
10. The scikit-learn function `model.predict(X_test)` does which of the following?
- A) Trains the model on the test data
 - B) Uses the trained model to generate predictions for the test set inputs**
 - C) Calculates the F1-Score automatically
 - D) Splits the test set into validation and holdout sets

11. Which combination of tools represents the standard Python data science stack?

A) Excel + R + MATLAB

B) Java + MySQL + Tableau

C) pandas + NumPy + matplotlib + scikit-learn + Jupyter Notebook

D) PowerPoint + Word + Access

Section B: Short Answer Questions (SQs)

Topic 6.1: Introduction to Data Science

1. Define data science in your own words and explain why it is described as an "interdisciplinary" field.
 2. List the five main disciplines that combine to form data science and briefly explain the role of each.
 3. Describe the six stages of the data science workflow pipeline in the correct order.
 4. Explain what the "feedback loop" stage of the data science pipeline means and why it is important.
 5. What is the difference between data and knowledge in the context of data science?
 6. Briefly explain how John Snow's cholera map (1854) demonstrated the core principles of data science, even without modern technology.
 7. Identify two real-world domains where data science is widely applied and give one specific example from each.
 8. Why is "domain knowledge" considered an essential component of data science alongside technical skills?
-

Topic 6.2: Understanding Data

Subtopic: Structured vs. Unstructured Data

1. Define structured data and provide two examples from everyday life.
2. Define unstructured data and provide two examples that differ from those in your answer to the previous question.
3. Differentiate between structured and unstructured data using a comparison table covering: format, examples, ease of analysis, and typical storage systems.
4. Why is unstructured data considered harder to analyse than structured data? What types of specialised tools are needed?
5. A hospital stores patient records in a table with columns for Name, Age, Diagnosis, and Treatment. Classify this as structured or unstructured data and justify your answer.
6. Approximately what percentage of the world's data is unstructured? What does this imply for the field of data science?

Subtopic: Data Sources and Collection Methods

1. List four different sources from which data can be collected and give a brief example of each.
2. Differentiate between manual data collection and automatic data collection. Give one advantage of each.
3. Explain the difference between real-time data collection and batch data collection. Give a practical example of each.
4. What does the phrase "Garbage In, Garbage Out" mean in the context of data collection?

Subtopic: Data Storage and Management

1. Define data storage and name three different types of storage systems used in data science.
 2. Explain the "3-2-1 backup rule" for data management and why it is important.
 3. Define data integrity and explain why it is important in a data management system.
 4. List four common data quality problems found in real-world datasets and briefly describe each.
 5. What is "mean imputation" and in what situation would a data scientist use it?
 6. Approximately how much of a data scientist's project time is typically spent on data cleaning? What does this suggest about the importance of this step?
 7. Distinguish between a "data warehouse" and a "data lake" in terms of how data is stored and used.
-

Topic 6.3: Overview of Machine Learning

1. Define machine learning in one or two sentences and explain how it differs from traditional programming.
 2. In traditional programming, what is the relationship between rules, data, and output? How does this relationship change in machine learning?
 3. Describe the "chef" analogy used in the study guide to explain the difference between traditional programming and machine learning.
 4. Why is machine learning better suited than traditional programming for tasks like face recognition? Provide a specific reason.
 5. Explain what "learning from data" means in the context of machine learning and how the model improves over time.
 6. What is Arthur Samuel's contribution to the field of machine learning? Briefly describe what his program did.
 7. Give two examples of real-world problems that are too complex for traditional rule-based programming and explain why machine learning is a better approach.
-

Topic 6.4: Machine Learning Mechanisms

Subtopic: Supervised Learning

1. Define supervised machine learning and explain what "labelled data" means.
2. Explain the two main types of supervised learning tasks — classification and regression — and give one example of each.
3. Describe the step-by-step process by which a supervised learning model learns from labelled training data and reduces its error.
4. Why does a supervised learning model's accuracy generally improve as more high-quality labelled examples are added to the training data?
5. Differentiate between the terms "feature" and "label" in the context of supervised learning.
6. Give three real-world applications of supervised learning and briefly explain what type of task (classification or regression) each represents.

Subtopic: Unsupervised Learning

1. Define unsupervised machine learning and explain what makes it different from supervised learning.
2. Explain what "clustering" is in unsupervised learning and give one practical example of how it is used in business.
3. Why is unsupervised learning useful for "data exploration"? What can it discover that supervised learning cannot?
4. Describe the Lego brick analogy used in the study guide to explain unsupervised learning.
5. Give two real-world applications of unsupervised learning beyond customer segmentation.

Subtopic: Reinforcement Learning

1. Define reinforcement learning and explain the roles of the "agent," "environment," "reward," and "penalty."
 2. Describe the "dog training" analogy for reinforcement learning. What does each element (dog, trainer, treat, "no") represent in the ML context?
 3. Explain the reinforcement learning cycle: how does the agent improve its decisions over repeated interactions with the environment?
 4. Give two real-world applications of reinforcement learning and briefly explain how rewards and penalties work in each case.
 5. Construct a comparison table distinguishing supervised, unsupervised, and reinforcement learning across four characteristics: use of labels, goal, learning style, and one example.
-

Topic 6.5: Applications of Machine Learning

1. Describe two specific ways machine learning is used in healthcare to improve patient outcomes.
 2. Explain how machine learning is used for fraud detection in financial systems. How does it identify suspicious transactions?
 3. Describe two ways machine learning is transforming education and its impact on student performance.
 4. Explain how a recommendation engine (like those used by Netflix or YouTube) uses machine learning to suggest content to users.
 5. What was the Netflix Prize? What type of machine learning problem did it represent and what was its impact on the field?
 6. Identify three everyday applications of machine learning that most people use without realising it, and briefly describe what the machine learning model is doing in each case.
 7. How does machine learning in business make marketing strategies more effective? Give a specific example.
 8. Explain how machine learning can accelerate drug discovery compared to traditional methods.
-

Topic 6.6: Building a Machine Learning Model

Subtopic: Feature Selection

1. Define the terms "feature" and "target value" in the context of building a machine learning model.
2. In a student grade prediction model with columns: Student_ID, Study_Hours, Attendance, Quiz_Average, Shoe_Size, and Grade, identify which columns should be kept as features, which should be removed, and which is the target value. Justify your choices.
3. Explain why including irrelevant features in a machine learning model can reduce its accuracy.

Subtopic: Feature Engineering

1. Define feature engineering and explain why it is a necessary step before training a machine learning model.
2. Explain what One-Hot Encoding is and why it is needed. Provide a brief example showing a "City" column being encoded.
3. Explain why feature scaling is important when training certain machine learning models. What problem does it solve?
4. Describe the Min-Max Scaling formula and explain what range the scaled values fall within.

5. Differentiate between "mean imputation" and "dropping rows with missing values" as strategies for handling missing data. When would each be appropriate?
6. Give one example of "creating a new feature" from existing data. Explain why the new feature might be more useful than the original.

Subtopic: Train-Test Split

1. Explain the purpose of the train-test split and why it is a fundamental practice in machine learning.
2. Using the exam paper analogy, explain why a model must never see the test data during training.
3. Define "overfitting" and explain one symptom that reveals a model is overfitting.
4. Define "underfitting" and explain what causes it and how it can be resolved.
5. Explain how the `random_state` parameter in `train_test_split` affects reproducibility of results.
6. Explain K-Fold Cross Validation and describe one advantage it has over a single train-validation split.

Subtopic: Predictive Model Building

1. Explain the meaning of the equation: $\text{Exam Score} = 7.55 \times \text{Study_Hours} + 16.9$. What does each number represent?
 2. Define Mean Squared Error (MSE) and explain what a lower MSE value indicates about model performance.
 3. Explain what R-squared (R^2) measures in a regression model. What does an R^2 value of 0.97 mean?
 4. Describe the six steps involved in building and evaluating a complete machine learning model from raw data to final predictions.
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Topic 6.7: Model Evaluation and Performance Metrics

1. Define the confusion matrix and explain the meaning of the four cells: TP, TN, FP, and FN.
2. Explain Type I Error and Type II Error in the context of a machine learning classifier.
3. Define accuracy and write its formula using TP, TN, FP, and FN.
4. Explain why accuracy can be a misleading metric when evaluating a model trained on an imbalanced dataset. Provide a specific numeric example.
5. Define precision and write its formula. Explain what a high precision value means for a spam filter.

6. Define recall and write its formula. Explain what a low recall value means for a cancer detection model.
 7. Explain the precision-recall trade-off. Give one scenario where maximising precision is the priority and one where maximising recall is the priority.
 8. Define F1-Score and write its formula. Explain why it is called the "harmonic mean" of precision and recall.
 9. Given $TP=60$, $TN=30$, $FP=15$, $FN=10$, calculate Accuracy, Precision, Recall, and F1-Score. Show all working.
 10. Explain why F1-Score is more informative than accuracy alone for evaluating a disease detection model.
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Topic 6.8: Validation and Model Improvement

1. Explain the difference between a validation set and a test set. Why must the two be kept strictly separate?
 2. Describe the three-way data split (train / validation / test). What is the specific purpose of each portion?
 3. Why should hyperparameters never be tuned using the test set? What bias does this introduce?
 4. Define hyperparameters and give two examples from machine learning models studied in this chapter.
 5. Explain the "speaker volume" analogy for hyperparameter tuning. What do "too loud" and "too quiet" represent in machine learning terms?
 6. Explain what model robustness means and describe two signs that indicate a model is NOT robust.
 7. Describe the process of hyperparameter tuning: what is tested, what is used to evaluate each setting, and how is the best setting selected?
 8. Explain what regularisation is and how it helps prevent overfitting.
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Topic 6.9: Predictive Modeling and Causality

1. Define predictive modelling and explain what type of data it relies on.
2. Define causality in the context of data science and explain what a "cause-and-effect relationship" means.
3. Define correlation and explain how it differs from causation.

4. Explain the "ice cream and drowning" example. What does it demonstrate about the relationship between two correlated variables?
 5. Explain the concept of a "confounding variable" (also called a hidden third variable) using a real-world example.
 6. Describe three conditions that must be met to establish a causal relationship between two variables.
 7. Give one example from education where a model might find a strong correlation that does not imply causation. Explain what the actual causal variable might be.
 8. Explain why a predictive model can make accurate predictions without understanding the underlying causal mechanisms.
 9. Why is it dangerous to make policy decisions (e.g., banning a product, changing a school rule) based solely on correlations found by a machine learning model?
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Topic 6.10: Machine Learning Tools and Platforms

1. Identify three main tools used in data science and machine learning and briefly describe what each is best suited for.
 2. What is Microsoft Excel used for in data analysis? List two of its main capabilities and one significant limitation.
 3. Explain what the R programming language is and in what professional context it is most widely used.
 4. Describe two features of R that make it well suited for statistical data analysis.
 5. Explain what makes Python the most popular language for machine learning and data science.
 6. Name four key Python libraries used in data science and describe the primary function of each.
 7. Describe what Jupyter Notebook is and explain two advantages it offers over writing code in a standard text editor.
 8. Construct a comparison table comparing Excel, R, and Python across four criteria: ease of learning, best use case, ability to handle large datasets, and machine learning capability.
 9. Explain what the `StandardScaler` in scikit-learn does and why it is used before training a KNN model.
 10. In the Python KNN model example from the study guide, what does `n_neighbors=3` mean and how does it affect predictions?
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Section C: Long / Extensive Questions (LQs)

Topic 6.1: Introduction to Data Science

1. The Foundations of Data Science

Data science is described as one of the most important fields of the 21st century.

- a) Define data science precisely and explain why it is called an "interdisciplinary" field. Name and describe the role of each contributing discipline.
- b) Describe the complete data science workflow pipeline, covering all six stages in order. Explain what happens at each stage and how the stages connect.
- c) Using a real-world scenario of your choice (not the cholera map), demonstrate how the six-stage pipeline would be applied from raw data collection to a final insight or decision.
- d) Evaluate the importance of the feedback loop stage. Why is data science often described as a continuous, iterative process rather than a one-time project?

1. From Data to Knowledge: Historical Foundations

- a) Describe John Snow's cholera investigation of 1854 in detail. What data did he collect? What patterns did he discover? What decision did he take and what was the outcome?
- b) Analyse how Snow's methodology reflects the core stages of modern data science, even though computers did not exist in 1854.
- c) Discuss why "domain knowledge" (in Snow's case, medical and public health knowledge) was critical to making sense of his data. What might have happened if he had no medical background?
- d) Justify why data science should be taught to all students, not just future computer scientists, using Snow's example as evidence.

Topic 6.2: Understanding Data

1. Structured and Unstructured Data: Types, Sources, and Management

- a) Define structured data and unstructured data. Construct a detailed comparison table covering: format, examples from at least three domains (healthcare, education, business), tools needed for analysis, and typical storage systems.
- b) Describe four different methods of data collection and classify each as manual or automatic, and real-time or batch. Give a real-world example for each method.
- c) Explain the concept of data quality. List five common data quality problems found in real-world

datasets and describe how each problem should be handled before analysis.

d) Evaluate the statement: "Data cleaning is the most important step in a data science project." Do you agree? Justify your answer with reference to what happens if this step is skipped.

1. Data Storage, Management, and the Challenge of Scale

a) Describe four types of data storage systems (files, relational databases, cloud storage, data lakes) and explain the strengths and appropriate use case for each.

b) Explain the importance of data security, backup, and integrity in a large organisation. What could go wrong if these principles are neglected?

c) Discuss the challenges of managing unstructured data at scale (e.g., millions of social media posts, medical images, or video recordings). What specialised tools and approaches are needed?

d) Design a data storage and management plan for a school that wants to track student performance data over five years. Specify what data will be collected, how it will be stored, backed up, and secured.

Topic 6.3: Overview of Machine Learning

1. Traditional Programming vs. Machine Learning: A Comprehensive Comparison

a) Explain traditional programming in terms of its input-process-output model. Give a concrete example showing how a programmer would write rules for a grade classification system.

b) Explain the machine learning paradigm and describe how it inverts the traditional programming model. Give the same grade classification example implemented as a machine learning problem.

c) Construct a detailed comparison table between traditional programming and machine learning covering: who writes the rules, how changes are made, how well it handles complex problems, suitable use cases, and limitations.

d) Describe three specific real-world problems where machine learning is clearly superior to traditional programming. For each, explain why writing explicit rules would be impossible or impractical.

e) Discuss one scenario where traditional programming might still be the better choice over machine learning. Justify your reasoning.

Topic 6.4: Machine Learning Mechanisms

1. The Three Mechanisms of Machine Learning: Deep Comparative Analysis

a) Define supervised machine learning. Explain the concepts of labelled data, features, labels, classification, and regression. Give a real-world example dataset for each task type.

b) Define unsupervised machine learning. Explain how clustering works and why no correct output is provided. Give three real-world applications where unsupervised learning reveals

insights that would be difficult to find otherwise.

c) Define reinforcement learning. Describe the agent-environment-reward-penalty cycle in detail. Give two real-world examples and trace through the learning process for each.

d) Construct a comprehensive comparison table for all three mechanisms covering: use of labels, learning style, goal, mathematical basis, data requirements, key algorithms, and three real-world applications.

e) Analyse which mechanism (supervised, unsupervised, or reinforcement) would be most appropriate for each of the following scenarios, and justify your choice:

- A bank wants to detect fraudulent transactions automatically.
- A music app wants to group users with similar listening tastes.
- A company wants to train a robot arm to assemble electronic components.

1. Supervised Learning in Depth: Labelled Data, Classification, and Regression

a) Explain what labelled data is and describe the process by which a supervised learning model is trained. Include the concepts of error measurement and iterative adjustment.

b) Differentiate between classification and regression with at least three examples of each. Explain what type of output each produces and what real-world questions each answers.

c) Describe the student grade prediction example in detail. What features would you use? What is the target value? What algorithm would be appropriate and why?

d) Discuss the limitations of supervised learning. What happens when labelled data is scarce, expensive to collect, or biased? Give a real-world example where biased labels led to a problematic model.

e) Evaluate the trade-off between model complexity and the amount of labelled training data needed. Why do more complex models generally require more data to avoid overfitting?

Topic 6.5: Applications of Machine Learning

1. Machine Learning Across Industries: Healthcare, Business, Education, and Daily Life

a) Describe in detail four specific applications of machine learning in healthcare. For each, explain what data is used, what type of ML task it represents, and what impact it has on patient care.

b) Describe four specific applications of machine learning in business. For each, explain the business problem being solved, the data input, and the measurable business benefit.

c) Describe three specific applications of machine learning in education. Evaluate both the potential benefits and the potential risks or ethical concerns associated with each.

d) Identify and describe five everyday machine learning applications that the average person uses. For each, identify whether it is primarily supervised, unsupervised, or reinforcement

learning.

e) Analyse the statement: "Machine learning is not replacing humans — it is augmenting human decision-making." Evaluate this claim using examples from at least two different sectors.

Topic 6.6: Building a Machine Learning Model

1. End-to-End Machine Learning Model Construction

- a) Describe the complete six-step process of building a machine learning model, from raw data to final prediction. For each step, explain what is done, why it is necessary, and what happens if it is skipped.
- b) Using the student grade prediction example, walk through each stage of the pipeline: define the features and target, describe the feature engineering steps needed, explain the train-test split decision, describe the model choice, and explain how predictions are made.
- c) Write complete, annotated Python code for the student grade prediction model using scikit-learn. The code should include: data preparation, feature selection, train-test split, model training with LinearRegression, prediction, and evaluation using MSE and R^2 .
- d) Interpret the model equation $\text{Exam Score} = 7.55 \times \text{Study_Hours} + 16.9$. Explain the meaning of the slope and intercept. Identify two limitations of this linear model in real-world deployment.
- e) Discuss what steps could be taken to improve the model's performance. Consider: adding more features, using a different algorithm, collecting more data, or applying feature engineering.

1. Feature Engineering: Transforming Raw Data for Machine Learning

- a) Define feature engineering and explain why raw data is almost never ready to be fed directly into a machine learning model.
- b) Describe and demonstrate four feature engineering techniques: (1) handling missing values using mean imputation, (2) One-Hot Encoding of a categorical column, (3) Min-Max Scaling of a numerical column, and (4) creating a new feature from existing columns. For each, provide a before-and-after data table showing the transformation.
- c) Explain the difference between feature selection and feature engineering. Why is it important to perform feature selection before feature engineering?
- d) Discuss the concept of the "curse of dimensionality" — what happens when a model has too many features relative to the number of training examples?
- e) Design a feature engineering pipeline for a house price prediction dataset with the following raw columns: House_ID, Bedrooms, Bathrooms, Total_Area_m2, House_Colour, Neighbourhood, Year_Built, Price. Specify every step you would take and justify each decision.

2. Train-Test Split, Overfitting, and Underfitting

- a) Explain the purpose of the train-test split and describe why the model must never see test data during training. Use the exam paper analogy in your explanation.
 - b) Define overfitting and underfitting. For each, describe: (1) the cause, (2) how it appears in training vs. test performance, (3) a real-world analogy, and (4) two methods to correct it.
 - c) Describe K-Fold Cross Validation in detail. Walk through a complete example with $K=5$, explaining what happens in each round and how the final performance estimate is calculated.
 - d) Discuss the trade-off in choosing the train-test split ratio. How does the choice of 70/30 vs. 80/20 vs. 90/10 affect model training and evaluation? Under what conditions would each ratio be appropriate?
 - e) A student builds a model that achieves 98% accuracy on training data but only 62% on test data. Diagnose the problem, identify likely causes, and propose a detailed correction plan.
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Topic 6.7: Model Evaluation and Performance Metrics

1. The Complete Evaluation Framework: Confusion Matrix, Accuracy, Precision, Recall, and F1-Score

- a) Draw and label a complete confusion matrix for a binary classifier. Define TP, TN, FP, and FN. Explain which cells represent correct predictions and which represent errors.
- b) Write and explain the formulas for Accuracy, Precision, Recall, and F1-Score. For each metric, describe what question it answers and when it is most useful.
- c) Given $TP=75$, $TN=120$, $FP=25$, $FN=30$, calculate all four metrics showing all steps clearly. Then write a paragraph interpreting these results in the context of a disease detection system.
- d) Explain the precision-recall trade-off with a detailed example. Design a scenario where maximising precision is the correct choice and another where maximising recall is the correct choice. Justify both decisions.
- e) Evaluate the following statement: "A model with 99% accuracy is always excellent." Using the imbalanced dataset example (99% healthy patients), prove why this statement is false and explain what metrics should be used instead.

2. Choosing the Right Metric: Applied Scenarios

- a) For each of the following scenarios, identify the most important evaluation metric (Precision, Recall, F1-Score, or Accuracy) and justify your choice:
 - A spam filter for a corporate email system
 - A medical test for detecting a rare but deadly virus
 - A system that recommends products to e-commerce shoppers
 - An airport security scanner that detects weapons
 - A loan approval system at a bank
- b) Explain why using only accuracy for an imbalanced dataset with 95% majority class and 5% minority class would be problematic. Calculate the accuracy of a model that always predicts

the majority class.

c) Construct a scenario where a model has high precision but very low recall, and another where a model has high recall but very low precision. For each, describe what the confusion matrix looks like and what real-world consequence each error type would have.

d) Discuss how the choice of evaluation metric reflects the values and priorities of the organisation deploying the model. Give two examples where different organisations solving the same problem might prioritise different metrics.

Topic 6.8: Validation and Model Improvement

1. Validation, Robustness, and Hyperparameter Tuning

a) Explain the three-way data split (training, validation, test) in detail. What is the purpose of each portion? Why must the test set never be used for tuning?

b) Define hyperparameters. Give three examples of hyperparameters from different ML models. Explain how each hyperparameter affects the model's learning behaviour.

c) Describe the complete process of hyperparameter tuning using the validation set. Walk through an example of tuning `max_depth` for a Decision Tree, showing at least five values tested and how the best value is selected.

d) Using the speaker volume analogy, explain overfitting (too loud) and underfitting (too quiet) in terms of hyperparameter settings. What is the "just right" zone called and how do we find it?

e) Explain K-Fold Cross Validation and discuss why it provides more reliable hyperparameter tuning results than a single validation split. Under what conditions is it especially valuable?

f) Define model robustness and reliability. Describe four tests you could run to check whether a trained model is robust before deploying it to a real-world production environment.

Topic 6.9: Predictive Modeling and Causality

1. Prediction vs. Causation: The Critical Distinction in Data Science

a) Define predictive modelling and explain how it uses historical patterns to estimate future outcomes. Give three examples from different domains (healthcare, education, business).

b) Define causality and explain what a cause-and-effect relationship requires. Describe three conditions that must be met to establish that variable A causes variable B.

c) Explain the difference between correlation and causation using two detailed examples: (1) the ice cream and drowning example, and (2) an example from education. For each, identify the hidden confounding variable.

d) Explain what a Randomised Control Trial (RCT) is and why it is the gold standard for establishing causation. Design a simple RCT to test whether additional study hours actually

cause better exam performance.

e) Discuss the real-world dangers of confusing correlation with causation in data-driven decision-making. Give two examples (one from business and one from healthcare) where acting on a correlation as if it were a cause could lead to harmful outcomes.

f) Evaluate the statement: "A machine learning model does not need to understand causality to be useful." Under what conditions is this true? Under what conditions is this dangerous?

Topic 6.10: Machine Learning Tools and Platforms

1. Choosing and Using the Right Tool: Excel, R, and Python

a) Compare Excel, R, and Python across the following dimensions: ease of use for beginners, statistical analysis capability, machine learning capability, scalability with large datasets, and industry adoption. Present your comparison in a structured table.

b) Describe five key Python libraries for data science (pandas, NumPy, matplotlib, seaborn, scikit-learn). For each, explain its primary function and give one concrete example of how it is used in a machine learning project.

c) Explain what Jupyter Notebook is and describe its structure (cells, markdown, code, output). Discuss three advantages it offers over writing Python code in a standard text file.

d) Walk through a complete Python machine learning workflow using scikit-learn for the student grade prediction problem. Include annotated code for: importing libraries, preparing data, splitting data, scaling features, training a KNN classifier, making predictions, and evaluating with accuracy, precision, recall, and F1-score.

e) Discuss the scenarios in which a data scientist might choose R over Python, and vice versa. Give three specific examples of tasks where R is preferable and three where Python is preferable.

2. Designing a Complete Data Science Solution: Capstone Case Study

A school district wants to build a machine learning system to predict which students are at risk of dropping out before completing their final year, so that counsellors can intervene early.

a) Identify and describe at least six features (input variables) that should be collected for this system. For each, explain why it is relevant to the prediction task.

b) Specify the target variable, the type of ML task (classification or regression), and the most appropriate ML mechanism (supervised, unsupervised, reinforcement). Justify each choice.

c) Describe the complete data preparation pipeline: how would you handle missing values, encode categorical variables, scale numerical features, and perform the train-test split?

d) Explain which evaluation metric (accuracy, precision, recall, or F1-score) is most critical for this system. Would you prioritise catching all at-risk students (high recall) or ensuring every student flagged is truly at-risk (high precision)? Justify your ethical reasoning.

e) Discuss three potential ethical concerns with deploying this system in a real school,

including privacy, bias, and the risk of labelling students unfairly. Propose one safeguard for each concern.

f) Suggest how the system could be improved over time using the feedback loop, hyperparameter tuning, and the addition of new data sources.

Cross-Topic and Synthesis Questions

1. Data Science in the Modern World: Synthesis and Evaluation

- a) Trace the full journey from "raw data" to a "deployed machine learning model" that predicts student exam performance. Cover every stage: data collection, storage, cleaning, feature engineering, model training, evaluation, validation, tuning, and deployment. Explain the role of each stage and the risks of skipping any step.
- b) Compare and contrast the three machine learning mechanisms (supervised, unsupervised, reinforcement) in terms of data requirements, learning approach, and appropriate application domains. Use a comparative table and at least three examples per mechanism.
- c) Discuss the statement: "Accuracy is the enemy of understanding in machine learning evaluation." What does this mean? Use precision, recall, F1-score, and the confusion matrix to explain why relying solely on accuracy can lead to dangerous conclusions in real-world deployments.
- d) Analyse the ethical implications of machine learning at scale. Consider: algorithmic bias (where training data reflects historical discrimination), privacy (sensitive personal data used for training), accountability (who is responsible when a model makes a harmful decision), and transparency (can users understand why a model decided what it did)?
- e) Evaluate the future of machine learning tools. How might tools like Python, Jupyter Notebook, and scikit-learn evolve over the next decade? What new challenges (data privacy laws, explainability requirements, real-time processing) will data scientists need to address?

2. Causality, Ethics, and Responsible Data Science

- a) Define and differentiate: correlation, prediction, and causation. Provide one original example (not from the study guide) for each concept that demonstrates the difference clearly.
- b) Explain why the difference between correlation and causation is especially critical in healthcare and education. Give one example in each domain where confusing the two could cause real harm.
- c) Describe what a confounding variable is and explain how it can make two unrelated variables appear strongly correlated. Give two original examples.
- d) Discuss the concept of "explainability" in machine learning — the ability for a model to explain why it made a specific prediction. Why is this especially important in healthcare, finance, and criminal justice? What is the trade-off between model accuracy and model explainability?

e) Evaluate the role of the data scientist as an ethical practitioner. What responsibilities do data scientists have when their models affect real people's lives (e.g., loan approvals, medical diagnoses, school admission decisions)? What principles should guide their work?

End of Comprehensive Question Bank — Chapter 6: Data and Analysis

Grade 12 Computer Science | MCQs: 190 | Short Questions: 112 | Long Questions: 19 (with multi-part sub-questions)

Teacher's Note on Usage

This question bank provides full coverage of every concept, definition, comparison, formula, code example, analogy, and application discussed in Chapter 6. Questions are distributed across all six levels of Bloom's Taxonomy:

- **Knowledge & Recall** — definitions, formulas, examples (MCQs 1–50)
- **Comprehension** — explanations and distinctions (MCQs 51–120; SQs 1–60)
- **Application** — calculations and code interpretation (MCQs 121–190; SQs 61–112)
- **Analysis** — comparisons and case interpretation (LQs 1–10)
- **Synthesis** — designing systems and pipelines (LQs 10–17)
- **Evaluation** — critical judgement and ethical reasoning (LQs 17–19)